

MAST7865 MSc Data Science Project Plan

Project Title:	Biomedical Knowledge Graph Construction Using Large Language Models: Evaluating Generalisation Beyond Benchmark Datasets
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1 Research Question(s)

Research Question 1: How effectively can an LLM-based information extraction pipeline extract biomedical entities and relationships from scientific abstracts for automatic knowledge graph construction?

Research Question 2: To what extent does the extraction pipeline's performance, tuned on an existing benchmark corpus, transfer to a newly assembled, contemporary biomedical corpus?

2 Overall Aim and Background

2.1 Overall Aim

The overall aim of this project is to develop and evaluate an LLM-based information extraction pipeline for automatic construction of biomedical knowledge graphs from scientific abstracts. The pipeline will be first developed and evaluated using an established biomedical benchmark corpus before being applied, without further refinement, to a contemporary corpus automatically retrieved from PubMed. This enables investigation of the extent to which benchmark-developed extraction pipelines generalise to current biomedical literature.

2.2 Background Motivation

2.2.1 Biomedical Literature Problem

The volume of new biomedical literature each year makes comprehensive manual on a large scale effectively impossible [1, 2]. On PubMed alone, hundreds of thousands of new articles are published monthly, with scientifically important findings commonly going unread due to falling outside the first page of relevant results - fewer than 20% of PubMed articles past the top 20 results are ever reviewed [3]. Automated extraction provides researchers a way to structure findings and important information at scale, however the value of such an extraction system depends heavily on whether the subdomains and key concepts are actually relevant to a research area and are not a fixed, inflexible historical sample. This provided support for a key design decision for this project: evaluating not only how well an extraction pipeline performs, but how well its performance generalises to contemporary, purpose-retrieved abstracts, rather than a fixed corpus, reliant on updates and changes that are separate from the main research process.

2.2.2 Knowledge Graphs

Knowledge graphs address the scale problem by organising extracted biomedical entities, such as genes, diseases, drugs and proteins, and the relationships between them into a structured, queryable form. This allows simpler and quicker human exploration of literature and can support clinical decision-making.

2.2.3 Existing Work

Recent literature has demonstrated that LLMs can perform relationship extraction on biomedical literature with competitive performance [4], driving a shift towards LLM-based automated knowledge graph construction [5]. However, much of the literature evaluates performance on held-out test splits of fixed benchmark datasets such as

BioRED [6] and BC5CDR [7]. These datasets are commonly used for evaluation in recent entity and relationship extraction studies [8, 9], but may not accurately reflect the continually evolving and growing state of biomedical literature.

2.2.4 Research Gap

As a result of the fact most studies use in-distribution evaluation, there is a meaningful gap in the literature. While benchmark datasets remain fixed, biomedical science is continuously growing and evolving - thus, entities and relationships to be extracted shift in ways that a fixed dataset does not reflect. Automated retrieval through assembling a corpus directly from current literature through a defined set of queries addresses this by allowing researchers to select precisely the domain, timeframe and scope required, rather than using the scope of an existing benchmark dataset that may have been constructed years earlier. Additionally, it is unclear how well these extraction systems built on a benchmark corpora generalise to newly published biomedical literature drawn from a different distribution within the same domain.

2.3 Data

2.3.1 Corpora Overview

Two corpora will be used in this project, corresponding to the two research questions. First, an established biomedical benchmark corpus (BioRED) will be used to develop and evaluate the extraction pipeline under RQ1, providing verified, pre-annotated entities and relationships to allow direct evaluation of extraction accuracy.

Second, a contemporary corpus will be assembled specifically for this project to address RQ2. Biomedical abstracts will be retrieved from PubMed using the NCBI Entrez API (via the `Biopython Entrez` module), with defined title/abstract search queries and publication year ranges, allowing the corpus to be constructed automatically for a defined biomedical subdomain rather than inherited from an existing dataset’s fixed scope. The contemporary corpus will be built from cardiology-related literature, retrieved across seven conditions (i.e., queries; atrial fibrillation, heart failure, coronary artery disease, myocardial infarction, hypertension, cardiomyopathy and heart valve disease) for publication years 2020-2026. As multiple queries can return the same citation, duplicate studies will be dropped given matching PubMed Identifiers (PMIDs). Citations with missing abstracts or PMIDs will also be dropped.

2.3.2 Variables Overview

Each record in the contemporary corpus contains the article’s PMID, title, abstract text, author list, journal name, publication year, DOI, and the search query used to retrieve the record. The BioRED benchmark corpus provides gold-standard entity and relationship annotations. This corpus includes more information than the contemporary corpus, however, for each record, PMID, title, abstract text, entity annotations, types and identifiers and relationship annotations and types will be used.

2.3.3 Limitations

Several potential limitations have been identified. Both corpora contain abstracts only and not the full citation body, which limits extraction to information reported in the abstract and may miss detail included in the body of a paper. Additionally, some records have incomplete data (such as missing DOIs). Publication years are also not evenly represented - number of papers returned is capped per query, therefore, more recent years with more publications may be overrepresented relative to earlier years. The NCBI E-utilities API also imposes rate limits, which constrains how quickly the corpus can be expanded or refreshed - this is not as restrictive in an exploratory study, but may become an issue if implemented at scale. Finally, as the contemporary corpus is newly assembled, it lacks expert-written entity and relationship annotations, although this will be addressed by manually annotating a representative subsample of the corpus, which will act as the ground truth for evaluating the pipeline’s performance under RQ2.

3 Objectives

The following objectives define the major stages required to achieve the project’s overall aim and answer the research questions.

- **Objective 1:** Develop an LLM-based biomedical information extraction pipeline capable of extracting entities and relationships from scientific abstracts and creating structured outputs suitable for knowledge graph construction.
- **Objective 2:** Evaluate and refine the extraction pipeline using the BioRED benchmark corpus through evaluation of entity and relationship extraction performance.

- **Objective 3:** Assemble a contemporary biomedical corpus from PubMed using the NCBI Entrez API, applying predefined search queries and preprocessing steps to produce a high-quality collection of scientific abstracts.
- **Objective 4:** Apply the frozen extraction pipeline to the contemporary biomedical corpus to automatically construct a biomedical knowledge graph.
- **Objective 5:** Manually annotate a representative subset of the contemporary corpus and compare extraction performance against the benchmark evaluation to investigate the system’s ability to generalise to contemporary biomedical literature.

4 Methodology

This project will be implemented in Python using an LLM-based information extraction pipeline. The methodology consists of two stages corresponding to the two research questions. The first stage focuses on and evaluates the extraction pipeline using the benchmark corpus, while the second explores its ability to generalise to a contemporary biomedical corpus assembled automatically from PubMed.

4.1 Benchmark Pipeline Development

An LLM will be used to extract entities and relationships from abstracts from the BioRED benchmark corpus, with outputs processed into a format suitable for knowledge graph construction.

Prompting will follow a few-shot approach, giving the model contextual example annotations drawn from the BioRED training set. This choice is supported by evidence from similar extraction domains - Martin et al. [10] observed that few-shot prompting significantly outperformed zero-shot for LLM-based entity and relationship extraction from cloud infrastructure logs, with one-shot prompting also showing a marginal increase in performance over zero-shot. While this performance is shown in a non-biomedical domain, it suggests use of few-shot prompting as the starting point for pipeline development, with the iterative refinement approach allowing this prompting choice to be changed.

Development will be carried out iteratively, tuning prompts and refining the extraction pipeline, followed by evaluation of extraction performance. The pipeline will be frozen following benchmark evaluation, so that its subsequent application to the contemporary corpus reflects performance under fixed, benchmark-tuned conditions.

4.2 Contemporary Corpus Construction

The contemporary corpus will be constructed according to the retrieval approach described in Section 2.3. Preprocessing will be applied to retrieved records including appropriate removal (by PMID) and data cleaning, outputting a complete, structured collection of abstracts. This process will be fully automated and reproducible, allowing the corpus to be reconstructed or extended if required.

Exploratory experiments have demonstrated the feasibility of the retrieval and preprocessing approach for the contemporary corpus.

4.3 Knowledge Graph Construction

Entities and relationships extracted by the pipeline will be mapped directly onto a graph structure: extracted entities as nodes and extracted relationships as edges. The resulting graph will be stored in a graph database (such as Neo4j, to be confirmed during implementation). This will support basic querying, enabling structured exploration of the extracted biomedical knowledge.

4.4 Evaluation Strategy

Extraction performance on the BioRED corpus will be evaluated using standard information extraction metrics (such as precision, recall, and F1) for both entity and relationship extraction. This will be benchmarked against BioRED’s gold-standard annotations to address RQ1. To evaluate generalisation to contemporary literature (RQ2), a representative subsample of the contemporary corpus will be manually annotated to serve as ground truth, following an annotation format consistent with BioRED. The frozen pipeline’s performance on this subsample will then be computed using the same metrics and compared to benchmark performance, this will allow evaluation of the extraction system’s ability to generalise to an out-of-distribution sample corpus.

4.5 Software and Technologies

The implementation will primarily use Python. The main libraries are expected to include Biopython, pandas, Neo4j (or a similar graph database), LLM APIs and evaluation libraries.

5 Timetable

Figure 1 presents the proposed project timeline, outlining the major phases of the project, key milestones and dependencies between tasks.

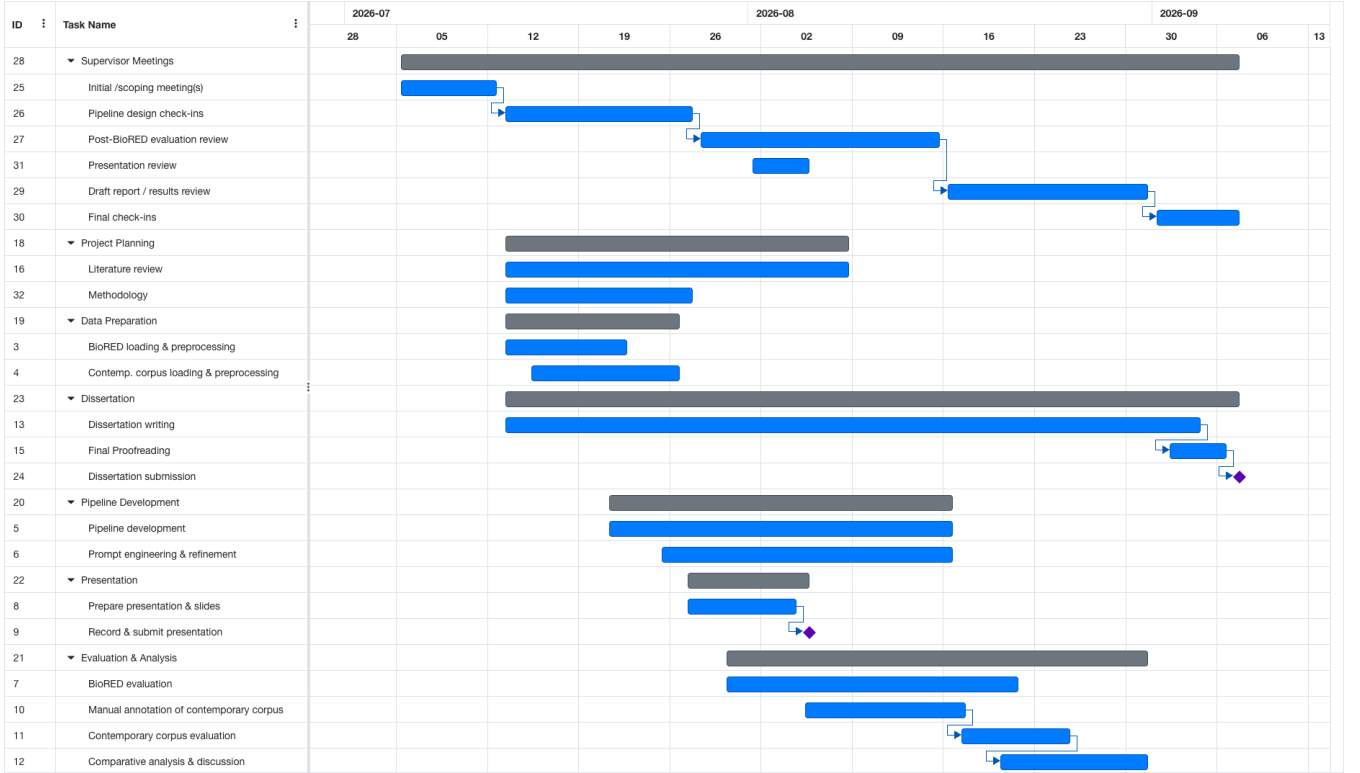


Figure 1: Proposed project timeline.

5.1 Project Schedule

The project timeline begins with literature review and methodological development, followed by data preparation and implementation of the information extraction pipeline. Benchmark evaluation is performed before the pipeline is frozen and applied to the contemporary biomedical corpus.

Following preparation of the recorded presentation, the later stages of the project focus on evaluating generalisation, analysing the results and writing the dissertation. Dissertation writing is scheduled throughout the project, allowing frequent refinement following supervisor feedback and mitigating risk of inadequate time allocation.

5.2 Risks and Mitigation

Table 1: Risks and mitigation strategies.

Risk	Mitigation
Missing or inconsistent PubMed metadata	Automated preprocessing and validation.
LLM prompt development takes longer than expected.	Freeze the pipeline after benchmark evaluation and avoid further refinement.
Manual annotation more time-consuming than expected	Restrict annotation to a representative subset.
Dissertation writing overruns	Write continuously throughout the project.
Computational limitations	Use abstracts rather than full-text articles.

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